

CWA SCIENCE CLASSES

Concept With Abhishek | Class 10 — Physics

Chapter 1: Light — Reflection and Refraction

1. Reflection of Light: Bouncing back of light when it falls on a polished/shiny surface. Laws: angle of incidence = angle of reflection; incident ray, reflected ray & normal lie in the same plane.
2. Spherical Mirrors: Concave mirror (converging), Convex mirror (diverging). Mirror formula: $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$.
3. Magnification (mirror) $m = -v/u = h'/h$.
4. Refraction of Light: Bending of light when it travels from one transparent medium to another due to change in speed.
5. Laws of Refraction: incident ray, refracted ray & normal lie in the same plane; Snell's Law $n_1 \sin(i) = n_2 \sin(r)$.
6. Lens Formula: $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$. Power of lens $P = 1/f$ (in metres), unit Dioptre (D).
7. Convex lens converges light rays and can form real or virtual images; concave lens always forms virtual, erect, diminished image.